CV Team Project: Happywhale - Whale and Dolphin Identification

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GitHub Link: https://github.com/fangfangirl/CV_final

1 Introduction

1.1 Problem Statement

Goal. The goal of Happywhale whale and dolphin identification is to recognize individual marine animals from images. Given an input image of a whale or dolphin, the model needs to predict the top 5 most likely individual identities. Unlike standard image classification, the test image may belong to either a known individual in the training set or a previously unseen individual. Therefore, the system must solve both identity recognition and open-set detection by deciding when to predict `new_individual`.

Task definition. Formally, given a query image x, the model predicts a ranked list of five candidate identities:

$$\hat{Y}(x) = [\hat{y}_1, \hat{y}_2, \hat{y}_3, \hat{y}_4, \hat{y}_5],$$

where each $\hat{y}_i$ is either one of the known training individuals or the special label `new_individual`. The prediction is evaluated by whether the correct identity appears in the top-5 list and how highly it is ranked.

1.2 Importance of the Problem

Why whale and dolphin identification?

- **Supports marine conservation and ecological research.** Individual whale and dolphin identification allows researchers to track animals over time, estimate population status, study migration patterns, and understand long-term behavioral changes.
- **Manual photo-identification is expensive and time-consuming.** In traditional marine mammal research, experts manually compare photographs based on natural markings such as tails, dorsal fins, heads, scars, and body patterns. This process requires thousands of hours of human effort and may limit the scale of scientific studies.
- **Automated identification can greatly reduce human effort.** A fast and accurate recognition system can reduce image identification time by over 99%, making large-scale marine mammal monitoring more practical and affordable.

1.3 The motivation or difficulty you address

Although photo-identification is useful for marine mammal research, most identification processes still rely heavily on manual matching by experts. As the image database becomes larger, manually comparing natural markings across many images becomes increasingly time-consuming and difficult to scale. Therefore, the main motivation of this project is to build an automatic identification system that can reduce manual effort and support large-scale whale and dolphin monitoring.

However, Happywhale is not a simple image classification problem. The main challenges are as follows:

- **Fine-grained individual differences.** Different whales or dolphins may have very similar appearances. The most discriminative cues often appear only in small local regions, such as dorsal fin shapes, scars, body textures, or natural markings.
- **Complex real-world images.** The images may contain different viewpoints, lighting conditions, image resolutions, occlusions, water splashes, partial bodies, and complex ocean backgrounds. These factors make it difficult for the model to focus on the truly informative regions.
- **Large-scale and imbalanced identities.** The dataset contains a large number of individual identities, and the number of images for each individual is highly imbalanced. Some individuals have many training images, while others only appear a few times.
- **Open-set recognition.** Some test images may belong to individuals that never appear in the training set. Therefore, the model must not only recognize known individuals but also decide when to predict `new_individual`.

To address these challenges, we formulate Happywhale identification as an embedding-based retrieval problem rather than a pure closed-set classification problem. Instead of directly forcing each query image into one of the known training identities, we learn a discriminative feature space in which images of the same individual are close to each other and images of different individuals are separated. During inference, each query image is compared with the training images through K-nearest neighbor retrieval. A similarity threshold is then used to decide whether the retrieved identities are reliable enough or whether `new_individual` should be inserted into the top-5 prediction list.

2 Related Work

2.1 Traditional Cetacean Photo-identification

Traditional cetacean photo-identification recognizes individual whales and dolphins using naturally occurring markings in photographs. Researchers commonly compare visual cues such as dorsal fin shapes, tail flukes, scars, notches, pigmentation patterns, and lateral body markings. This approach has been widely used in marine mammal studies because it enables non-invasive tracking of individuals over time and supports population monitoring [Würsig and Jefferson, 1990, Urian et al., 2015].

For different cetacean species, different body regions may provide the most reliable identity cues. For example, dolphin photo-identification often relies on dorsal fin outlines and marks, while humpback whale identification commonly uses tail fluke patterns together with dorsal-fin and lateral body markings [Mazzoil et al., 2004, Friday et al., 2000]. Previous studies have also analyzed natural marking patterns for common minke whales and white-beaked dolphins, showing that fin outlines, notches, injury marks, and other persistent markings can be useful for individual recognition [Bertulli et al., 2016].

Pros. Traditional photo-identification is interpretable, non-invasive, and follows the workflow used by marine mammal researchers.

Cons. It requires substantial manual effort and becomes difficult to scale when the number of images and individuals increases. It can also suffer from matching errors when image quality is low or when individuals have subtle or temporary markings.

2.2 Local and Species-specific Matching Methods

Before deep learning became dominant, many animal re-identification systems relied on local descriptor matching. These methods detect distinctive keypoints or local patterns from an image and compare them with a database of known individuals. Typical examples include SIFT-, SURF-, ORB-, SuperPoint-, or HotSpotter-like matching methods. The identity prediction is usually made by retrieving the database individual with the strongest local descriptor correspondence [Čermák et al., 2024].

Local descriptor methods are attractive because they are often plug-and-play and do not require task-specific fine-tuning. This makes them useful for ecological applications where researchers may not have enough labeled data or computational resources to train deep neural networks. However, these methods can be less effective when images contain strong viewpoint changes, background clutter, low resolution, or large pose variation. They may also scale poorly when the database contains many images and identities.

Another line of work uses species-specific matching methods. These methods are designed for a particular species or a group of visually similar species. For example, some methods focus on special patterns such as spots, stripes, whisker marks, or aligned body patches. For cetaceans, this idea is related to using body regions such as dorsal fins, tail flukes, scars, or lateral markings for identification. While species-specific methods can be effective when the target visual patterns are stable and distinctive, they are difficult to transfer to other species or datasets.

Pros. Local and species-specific methods can be interpretable and may work well when the target species has stable and distinctive visual markings. They may also require less training data than deep learning methods.

Cons. Their performance often depends on handcrafted assumptions, image quality, and accurate alignment or region selection. They are also less transferable across species and may not scale well to large, diverse animal re-identification datasets.

2.3 Deep Descriptor and Margin-based Identity Learning

With the progress of deep learning, animal re-identification is increasingly formulated as a deep descriptor learning problem. A neural network is trained to map each image into a compact embedding vector, and visually similar images are expected to have nearby embeddings. A common training strategy is to treat each known individual as a separate identity class and train the model with identity-level supervision.

A straightforward approach is ordinary softmax classification. Given an input image, the classifier predicts a probability distribution over all known identities. This formulation is simple and effective when all test identities are already included in the training set. However, ordinary softmax classification does not explicitly enforce large separation between visually similar identities. This is challenging for whale and dolphin photo-identification, where different individuals may look similar, while images of the same individual may vary due to viewpoint, lighting, pose, occlusion, crop quality, and visible body regions.

Margin-based classification losses address this limitation by making the learned embedding space more discriminative. ArcFace introduces an additive angular margin in the normalized feature space [Deng et al., 2019]. Instead of only maximizing the probability of the correct identity, ArcFace requires the model to classify the image correctly with an additional angular margin. This encourages samples from the same identity to be more compact and pushes samples from different identities farther apart. In cetacean photo-identification, Patton et al. applied ArcFace-based classification heads to jointly learn species and individual identities in a multi-species setting [Patton et al., 2023].

Sub-center ArcFace further extends ArcFace by assigning multiple sub-centers to each class [Deng et al., 2020]. Instead of forcing all samples of the same identity to fit a single class center, each sample only needs to be close to one of its positive sub-centers. This design is useful for animal re-identification because the same individual may appear under different viewpoints, body regions, crop types, or image qualities.

Another difficulty is the long-tailed distribution of identity labels. In large-scale animal identification datasets, some individuals have many images, while many others only have a few examples. This imbalance can cause the classifier to be dominated by frequent identities. Prior work on imbalanced learning, such as LDAM and Class-Balanced Loss, shows that class frequency should be considered when designing the training objective [Cao et al., 2019, Cui et al., 2019]. Inspired by these ideas, adaptive-margin classification heads can adjust margins according to class frequency, encouraging the model to learn better representations for rare identities.

Pros. Deep descriptors and margin-based identity learning can produce discriminative embeddings for fine-grained animal re-identification. ArcFace improves angular separation between identities, Sub-center ArcFace handles intra-class visual variation, and adaptive margins address long-tailed identity distributions.

Cons. These methods are trained using known identity labels. If the final prediction only relies on a fixed classification head, the model cannot naturally handle unseen individuals.

2.4 Embedding-based Retrieval and Open-set Identification

Although classification-based identity learning can produce discriminative features, a fixed classification head is limited to the identities observed during training. In open-set animal photo-identification, a query image may belong to either a known individual in the reference database or a previously unseen individual. Therefore, forcing every query image to be classified into one of the training identities can lead to incorrect predictions, especially when the correct answer should be an unseen individual.

To address this limitation, many photo-identification and re-identification systems use embedding-based retrieval. Instead of directly relying on the output probability distribution of the classification head, the trained network is used as a feature extractor. Each image is mapped into an embedding vector, and identity prediction is performed by comparing the query embedding with a gallery of known individual embeddings. The most similar gallery images are retrieved using nearest-neighbor search, commonly based on cosine similarity or Euclidean distance. The identities of the top-ranked neighbors are then used as candidate predictions. This formulation is closely related to deep metric learning, where the model is trained to make images of the same individual close to each other and images of different individuals far apart [Miele et al., 2021].

This retrieval formulation is more suitable for open-set recognition because it does not require the model to assign every query image to a fixed classifier class. If a query image is close to some known individual embeddings, the system can predict that known identity. If the maximum similarity is lower than a threshold, the image can instead be treated as an unseen individual. Recent wildlife re-identification benchmarks and toolkits also adopt retrieval-based evaluation protocols, reflecting the importance of nearest-neighbor search in animal individual identification [Čermák et al., 2024].

Pros. Embedding-based retrieval is more flexible than a fixed classification head because it compares query images against known individuals without forcing every sample into a closed-set class. It is especially useful for open-set photo-identification, where unseen individuals may appear during testing. KNN retrieval also allows the system to rank multiple candidate identities and insert `new_individual` based on a similarity threshold.

Cons. Retrieval performance strongly depends on the quality of the learned embedding and the choice of similarity threshold. If the embedding space is not well separated, visually similar but different individuals may be confused. If the threshold is too high, known individuals may be incorrectly predicted as `new_individual`; if it is too low, unseen individuals may be forced into known identities.

2.5 Cropping and Species-aware Identification

In animal photo-identification, not all regions of an image are equally informative. For whales and dolphins, identity cues often appear in specific body parts, such as dorsal fins, tails, scars, lateral body markings, and texture patterns. Therefore, cropping important regions can reduce background noise and help the model focus on discriminative visual features. This idea is especially relevant to real-world marine animal images, which may contain ocean backgrounds, water splashes, partial bodies, or irrelevant regions. Comparing different image views, such as raw images, fullbody crops, and dorsal-fin or backfin crops, can help evaluate which visual regions are most useful for individual recognition.

Species information is another useful cue for multi-species animal identification. In addition to identity classification, species classification can be used as auxiliary supervision. Since species labels are coarser and less sparse than individual identity labels, they can provide a more stable training signal. A species-level auxiliary head encourages the shared backbone to learn biologically meaningful visual features, which may further support fine-grained identity classification. Patton et al. showed that species and individual identity can be jointly learned in a multi-species cetacean photo-identification model [Patton et al., 2023].

During inference, species information can also be used for species-aware re-ranking. For example, retrieval candidates with species labels inconsistent with the query prediction can be adjusted or penalized. This strategy may reduce cross-species retrieval errors, but it depends on the reliability of species prediction.

Pros. Cropping can reduce irrelevant background and focus the model on identity-related regions. Species-aware learning may reduce incorrect retrieval across different species and provide additional supervision for the feature extractor.

Cons. The benefit of cropping depends on crop quality. If the crop misses important markings or removes useful context, performance may decrease. Similarly, if species prediction is wrong, species-aware re-ranking may amplify errors rather than improve the final prediction.

2.6 Universal Animal Re-identification Models

Recent animal re-identification research has started to move from dataset-specific or species-specific models toward more universal feature extractors. Instead of training a separate model for each species or dataset, these methods aim to learn representations that can transfer across multiple animal re-identification tasks. This direction is motivated by the lack of standardization in dataset usage, evaluation protocols, and method design across wildlife re-identification studies [Čermák et al., 2024].

WildlifeDatasets is an open-source toolkit that standardizes access to many public wildlife re-identification datasets and supports preprocessing, dataset splitting, feature extraction, identity matching, evaluation, and model fine-tuning [Čermák et al., 2024]. The same work categorizes common wildlife re-identification approaches into local descriptors, deep descriptors, and species-specific methods. Based on large-scale experiments across many animal re-identification datasets, the authors further introduce MegaDescriptor, a Swin-based feature extractor trained for individual re-identification across a wide range of species. Their experiments show that MegaDescriptor outperforms general pretrained models such as ImageNet, CLIP, and DINOv2 on many animal re-identification datasets.

This trend is related to self-supervised visual representation learning. Models such as DINO and DINOv2 learn general image features without manually annotated labels and have shown strong transfer ability across different visual tasks [Caron et al., 2021, Oquab et al., 2023]. However, general-purpose visual representations are not always sufficient for animal individual recognition. For example, WildlifeDatasets reports that MegaDescriptor performs better than ImageNet, CLIP, and DINOv2 on the HappyWhale dataset, but the absolute performance remains much lower than on several other animal re-identification datasets [Čermák et al., 2024]. This suggests that large-scale whale and dolphin identification still requires task-specific adaptation.

In our setting, MegaDescriptor and DINO-based representations are explored as alternative feature extractors. However, directly applying these general representations does not consistently improve identification performance. This observation suggests that universal feature extractors are useful starting points, but task-specific fine-tuning, appropriate crop selection, and retrieval calibration remain important for fine-grained open-set whale and dolphin identification.

Pros. Universal animal re-identification models can reduce the need to train a separate model for each species or dataset. They provide transferable embeddings and can serve as strong starting points for new animal identification tasks.

Cons. They may still fail to capture subtle identity-specific cues required for difficult fine-grained tasks. For whale and dolphin photo-identification, task-specific adaptation, crop selection, and threshold calibration remain necessary.

3 Proposed Approach

3.1 Overview

Figure 1 shows the overview of our proposed pipeline. We formulate whale and dolphin individual identification as a multi-view embedding retrieval problem. Instead of relying only on a closed-set classifier, we first train an embedding model to learn discriminative identity representations, and then use retrieval-based inference to generate top-5 predictions.

The pipeline consists of four stages. First, we prepare multiple image views from each raw image, including raw view, fullbody crop, and backfin crop. These views provide different visual cues for individual identification. Second, we train embedding models using margin-based identity supervision and species auxiliary supervision. The identity head is based on Sub-center ArcFace with adaptive margin, while the species head provides coarse-grained auxiliary supervision.

Third, during inference, we extract embeddings and logits for both reference and query images. For each model-view pair, we compute KNN retrieval scores, identity logit scores, and prototype-based identity scores. These scores are fused into an identity score matrix, optionally adjusted by species-aware re-ranking, and then combined across different models and crop views. Finally, `new_individual` is inserted as a threshold-based candidate for top-5 prediction.

Fourth, we perform pseudo-label self-training by selecting high-confidence test predictions and adding them back into the training set for another round of fine-tuning.

3.2 Multi-view Image Preparation

Identity cues of whales and dolphins may appear in different body regions, such as dorsal fins, scars, lateral body markings, and body textures. However, raw images often contain ocean backgrounds, water splashes, partial bodies, or irrelevant regions. Therefore, we use a multi-view image preparation strategy to provide the model with complementary visual information.

For each image, we consider three types of views. The raw view preserves the original context and global appearance. The fullbody crop focuses on the visible animal body and reduces irrelevant background. The backfin crop emphasizes local fin-related cues, which are often important for individual identification. During training, these views are mixed as input images, so the model can learn robust features from different visual perspectives. During inference, predictions from different views can be combined to improve stability.

In the final ensemble, most models use three views, including fullbody, backfin, and raw images. For the EfficientNet-B6 branch, we additionally include a fullbody_charm view, which is treated as an additional fullbody-style crop during crop-level score fusion.

3.3 Embedding Model Training

Given an input image view, we use a visual backbone to extract feature maps. Our main backbones include EfficientNet-B6, EfficientNet-B7, EfficientNetV2-XL, and ConvNeXt-based models. The extracted feature maps are aggregated by Generalized Mean Pooling (GeM), followed by Batch Normalization and L2 normalization to obtain a compact embedding vector.

Given a feature map $X \in \mathbb{R}^{C\times H\times W}$, GeM pooling computes one pooled value for each channel:

$$f_c = \left(\frac{1}{|\Omega|} \sum_{u\in\Omega} X_{c,u}^p \right)^{\frac{1}{p}},$$

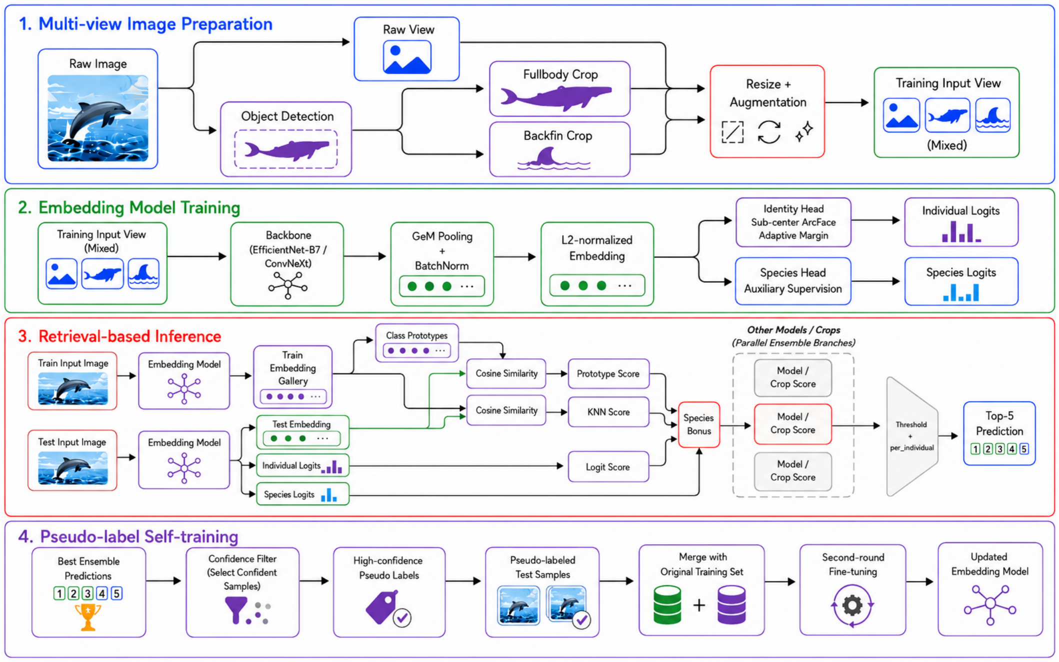


Figure 1: Overview of the proposed multi-view and multi-backbone retrieval pipeline. First, each image is prepared into multiple views, including raw, fullbody, and backfin crops. Second, an embedding model is trained with Sub-center ArcFace identity supervision and species auxiliary supervision. Third, during inference, embeddings and logits are converted into KNN retrieval scores, identity logit scores, and single-prototype scores. These scores are fused, adjusted by species-aware re-ranking, combined across model/crop branches, and then used with threshold-based `new_individual` insertion to generate the top-5 prediction. Fourth, high-confidence predictions are selected as pseudo labels and merged into the training set for further fine-tuning.

where $X_{c,u}$ is the activation of channel c at spatial location u, Ω denotes all spatial locations in the feature map, and p is a learnable pooling parameter. When $p = 1$, GeM becomes average pooling. When p becomes large, GeM behaves closer to max pooling. Therefore, GeM can emphasize discriminative local responses while still preserving global image-level information. This is useful for whale and dolphin identification because identity cues such as dorsal fin shapes, scars, and body markings may appear only in small local regions.

The pooled feature is then passed through Batch Normalization and L2 normalization:

$$z = \frac{\mathrm{BN}(f)}{\|\mathrm{BN}(f)\|_2},$$

where z is the final normalized image embedding. This embedding is used for both identity classification during training and nearest-neighbor retrieval during inference.

The normalized embedding is used for two supervision signals. The first is the individual identity head. We use Sub-center ArcFace with adaptive margin to learn discriminative identity embeddings. Sub-center ArcFace allows each identity to have multiple class centers, which is useful because the same whale or dolphin may appear under different viewpoints, crop types, lighting conditions, and body regions. The adaptive margin further accounts for the long-tailed distribution of identities, where some individuals have many training images while others only have a few.

The second supervision signal is the species auxiliary head. Species labels are coarser than individual identity labels and provide a more stable training signal. By jointly learning species and individual identity, the shared backbone can learn both species-level and individual-level visual cues. The total training loss is defined as:

$$\mathcal{L} = \mathcal{L}_{id} + \lambda_{sp}\mathcal{L}_{sp},$$

where $\mathcal{L}_{id}$ is the Sub-center ArcFace identity loss, $\mathcal{L}_{sp}$ is the species classification loss, and λ_{sp} controls the weight of species supervision.

3.4 Retrieval-based Inference

After training, we use the model as both an embedding extractor and a source of identity and species logits. For each crop view, we extract outputs for both reference images and query images.

The inference pipeline combines flip test-time augmentation, embedding-based KNN retrieval, identity logits, prototype-based identity scoring, species-aware score adjustment, crop ensemble, model ensemble, and `new_individual` thresholding.

Embedding extraction and flip test-time augmentation. For each image x, we perform inference on both the original image and its horizontally flipped version. Let $T_{\text{flip}}(x)$ denote the flipped image. The model produces an embedding and logits for both views:

$$(z^{orig}, l^{orig}, q^{orig}) = F(x),$$

$$(z^{flip}, l^{flip}, q^{flip}) = F(T_{\text{flip}}(x)),$$

where z is the identity embedding, l is the identity logit vector, and q is the species logit vector. For the classification outputs, we average the original and flipped logits:

$$l^{tta} = \frac{1}{2}\left(l^{orig} + l^{flip}\right),$$

$$q^{tta} = \frac{1}{2}\left(q^{orig} + q^{flip}\right).$$

For retrieval, we keep both original and flipped embeddings. The reference gallery is constructed by concatenating original and flipped embeddings of all training images:

$$\mathcal{G} = \{(z_i^{orig}, y_i)\}_{i=1}^{N} \cup \{(z_i^{flip}, y_i)\}_{i=1}^{N},$$

where y_i is the identity label of the i-th reference image. This allows the retrieval stage to use both original and flipped visual representations.

KNN identity retrieval. Given a query image x, we compute cosine similarity between the query embedding and all reference embeddings. Since all embeddings are L2-normalized, cosine similarity can be computed by an inner product:

$$\text{sim}(x, x_i) = z(x)^\top z(x_i).$$

For each query, we retrieve the top-K nearest reference images. The retrieved image-level scores are then converted into identity-level scores. For an identity class c, the KNN score is defined as the maximum similarity among retrieved reference images belonging to that identity:

$$S_{\text{knn}}^{orig}(c \mid x) = \max_{i \in \mathcal{N}_K(z^{orig}),\, y_i = c} z^{orig}(x)^\top z_i,$$

where $\mathcal{N}_K(z^{orig})$ denotes the top-K nearest reference embeddings retrieved by the original query embedding. The same process is also applied to the flipped query embedding:

$$S_{\text{knn}}^{flip}(c \mid x) = \max_{i \in \mathcal{N}_K(z^{flip}),\, y_i = c} z^{flip}(x)^\top z_i.$$

The final KNN score is the average of the original and flipped retrieval scores:

$$S_{\text{knn}}(c \mid x) = \frac{1}{2}\left(S_{\text{knn}}^{orig}(c \mid x) + S_{\text{knn}}^{flip}(c \mid x)\right).$$

KNN-logit-prototype score fusion. In addition to KNN retrieval, the identity classification head provides logit scores for known identities. Although the classifier output alone cannot naturally handle unseen individuals, it still contains useful class-level confidence learned during training. Let $S_{\text{logit}}(c \mid x)$ denote the restored identity logit score of class c from l^{tta}.

We also introduce a prototype-based identity score. The motivation is that KNN focuses on nearest reference examples, while prototypes provide a more global class-level representation. In the final ensemble, we use the single-prototype setting. For each identity class, one prototype is constructed by averaging the normalized reference embeddings belonging to that identity.

Let $\mathcal{I}_c$ denote the set of reference embeddings belonging to identity class c. The prototype of class c is defined as:

$$p_c = \frac{1}{|\mathcal{I}_c|} \sum_{i \in \mathcal{I}_c} z_i,$$

where z_i is the normalized embedding of the i-th reference image. The prototype is then L2-normalized:

$$\tilde{p}_c = \frac{p_c}{\|p_c\|_2}.$$

Given a query image x, the single-prototype score is computed by cosine similarity:

$$S_{\text{proto}}(c \mid x) = z(x)^\top \tilde{p}_c.$$

For flip test-time augmentation, prototype scores are computed for both the original and flipped query embeddings and then averaged:

$$S_{\text{proto}}(c \mid x) = \frac{1}{2}\left(S_{\text{proto}}(c \mid x^{orig}) + S_{\text{proto}}(c \mid x^{flip})\right).$$

Finally, we combine the KNN retrieval score, identity logit score, and prototype score:

$$S(c \mid x) = w_{\text{knn}} S_{\text{knn}}(c \mid x) + w_{\text{logit}} S_{\text{logit}}(c \mid x) + w_{\text{proto}} S_{\text{proto}}(c \mid x),$$

where w_{knn}, w_{logit}, and w_{proto} are fusion weights. In practice, the weights are normalized to sum to one:

$$w_{\text{knn}} + w_{\text{logit}} + w_{\text{proto}} = 1.$$

This fusion combines instance-level nearest-neighbor similarity, classifier-based identity confidence, and prototype-based class-level similarity.

We also considered a multi-prototype variant, where identities with enough samples are clustered into multiple centers. However, the final ensemble uses the single-prototype setting for simplicity and stability.

Species-aware score adjustment. After obtaining the fused identity score from KNN retrieval, identity logits, and prototype scoring, we further use the species head to adjust individual identity scores.

First, we build a mapping from each identity class to its species label using the training data. For each query image, the species head predicts the most likely species labels from q^{tta}. If a candidate identity belongs to one of the predicted species, we add a small bonus to its score:

$$S'(c \mid x) = S(c \mid x) + \sum_{r=1}^{K_s} \frac{\beta}{r} \cdot \mathbb{I}\left[s(c) = \hat{s}_r(x)\right],$$

where $s(c)$ is the species label of identity class c, $\hat{s}_r(x)$ is the r-th ranked predicted species of query image x, K_s is the number of predicted species used for adjustment, and β is the species bonus weight. In the simplest setting, only the top-1 predicted species is used. This adjustment is designed to reduce cross-species retrieval errors, but it depends on the reliability of species prediction.

Model and crop ensemble. The above inference process is performed independently for each trained model and each crop view. Each model-view pair produces a score matrix over all known identities. In our implementation, each score matrix can be built from multiple components, including KNN scores, identity logits, and prototype scores. Since these components are converted into the same identity-score matrix format, different model and crop outputs can be fused even when their embedding dimensions are different.

Different models and crop views may capture complementary information. For example, EfficientNet-B6/B7 and ConvNeXt-based models may learn different visual representations, while raw, fullbody, and backfin views focus on different regions of the animal.

The final ensemble score is computed by averaging or weighted averaging the score matrices from different models and crop views:

$$S_{\text{ens}}(c \mid x) = \sum_{m \in \mathcal{M}} \sum_{v \in \mathcal{V}} w_{m,v} S'_{m,v}(c \mid x),$$

where $\mathcal{M}$ denotes the set of models, $\mathcal{V}$ denotes the set of crop views, $S'_{m,v}(c \mid x)$ is the species-adjusted score of identity class c from model m and crop view v, and $w_{m,v}$ is the ensemble weight.

In practice, we evaluate multiple weight combinations and select the final setting according to validation performance and submission feedback. This allows the ensemble to assign higher weights to stronger model-view pairs and lower weights to less reliable ones. For example, if the fullbody view performs better than the raw or backfin views, the ensemble can place a larger weight on the fullbody score matrix. This strategy is more flexible than simple averaging and helps improve the robustness of the final prediction.

`new_individual` thresholding. Since the test set may contain individuals that do not appear in the training set, the model needs to decide when to predict `new_individual`. In our implementation, we do not insert `new_individual` at a fixed rank. Instead, we treat it as an additional candidate class with a fixed threshold score τ.

After obtaining the final ensemble scores for all known identities, we append the score τ to the score vector:

$$S_{\text{all}}(x) = [S_{\text{ens}}(1 \mid x), \ldots, S_{\text{ens}}(C \mid x), \tau],$$

where C is the number of known identity classes. Then, all known identities and `new_individual` are ranked together, and the final prediction is the top-5 candidates:

$$\hat{Y}(x) = \text{Top5}\left(S_{\text{all}}(x)\right).$$

Therefore, the position of `new_individual` depends on how its threshold score compares with the known-identity scores. If τ is higher than all known-identity scores, `new_individual` becomes the top-1 prediction. If τ is between the known-identity scores, it may appear at rank 2 to rank 5. If τ is lower than the top-5 known-identity scores, it will not appear in the final prediction.

The threshold τ is selected automatically by binary search. We first specify a target top-1 `new_individual` ratio, such as 0.165, and then search for a threshold that makes the proportion of top-1 `new_individual` predictions close to this target. This allows us to control how frequently the model predicts unseen individuals while still ranking `new_individual` naturally with all known identities.

3.5 Pseudo-label Self-training

After obtaining predictions from the strongest model or ensemble, we use pseudo-label self-training as the fourth stage of our pipeline. The goal is to use high-confidence test-domain predictions as additional training samples, so that the model can better adapt to the test image distribution.

For each test image, the ensemble produces a ranked list of candidate identities and their confidence scores. We select pseudo-labeled samples using conservative confidence filtering. First, the top-1 prediction must not be `new_individual`. Second, the top-1 confidence score must be higher than a predefined threshold. Third, the score margin between the top-1 and top-2 predictions must be sufficiently large, so that ambiguous samples are removed. This filtering strategy reduces the risk of adding noisy pseudo labels.

The selected pseudo-labeled dataset is denoted as:

$$\mathcal{D}_{pseudo} = \{(x_i, \hat{y}_i)\}_{i=1}^{M},$$

where x_i is a selected test image and $\hat{y}_i$ is its pseudo identity label. We then merge the pseudo-labeled samples with the original training set:

$$\mathcal{D}_{train}^{+} = \mathcal{D}_{train} \cup \mathcal{D}_{pseudo}.$$

We use two pseudo-label training strategies. The first strategy is **from-scratch pseudo training**. In this setting, the model is trained from the beginning using the expanded dataset $\mathcal{D}_{train}^{+}$. This allows the model to learn from the original labeled images and pseudo-labeled test-domain images throughout the entire training process.

The second strategy is **resume pseudo fine-tuning**. In this setting, we start from a model checkpoint that has already been trained on the original labeled dataset, such as the checkpoint around epoch 30. Then, we continue training the model using $\mathcal{D}_{train}^{+}$. This strategy is more efficient because the model has already learned a strong identity embedding space before pseudo labels are introduced. It also reduces the training cost compared with restarting the full training process.

Pseudo-label self-training can increase the amount of training data and expose the model to additional test-domain images. However, it also introduces the risk of confirmation bias. If incorrect predictions are added into the training set, the model may reinforce its own mistakes. Therefore, we apply confidence and margin-based filtering to keep only reliable pseudo labels.

3.6 Design Rationale

The design of our approach is motivated by the main challenges of Happywhale whale and dolphin identification. First, the task is a fine-grained recognition problem, where different individuals may have very similar appearances. Therefore, we use margin-based identity learning with Sub-center ArcFace to learn a more discriminative embedding space. The use of multiple sub-centers allows the model to better handle intra-individual variation caused by different viewpoints, crop types, lighting conditions, and visible body regions.

Second, the dataset contains a large number of identities with a highly imbalanced distribution. Some individuals have many training images, while many others only appear a few times. To address this issue, we apply an adaptive margin strategy in the identity head, so that the training objective can better reflect class-frequency imbalance.

Third, identity cues may appear in different image regions. Raw images preserve global context, fullbody crops focus on the animal body, and backfin crops emphasize local fin-related features. Therefore, we use multi-view image inputs and combine different crop views during inference. This design allows the model to use complementary visual information from different regions.

Fourth, the test set may contain individuals unseen during training. A closed-set classifier alone cannot naturally handle this open-set setting, so we use embedding-based KNN retrieval and treat `new_individual` as an additional threshold-based candidate during ranking. This makes the inference process more suitable for open-set individual identification.

Fifth, the labeled training set may not fully represent the visual distribution of the test set. Therefore, we use pseudo-label self-training to exploit high-confidence test-domain predictions. By adding reliable pseudo-labeled samples back into the training set, the model can learn from additional test-domain images and further adapt its embedding space. We use confidence and top-1/top-2 margin filtering to reduce noisy pseudo labels and avoid reinforcing incorrect predictions.

Finally, we use species auxiliary supervision and species-aware score adjustment to reduce cross-species confusion. We also ensemble multiple models and crop views to improve robustness.

4 Experimental Results

4.1 Dataset and Evaluation Metric

Dataset Metric. We evaluate our method on the Happywhale Whale and Dolphin Identification dataset. The goal of this task is to identify individual whales and dolphins from images. For each test image, the model is required to output a ranked list of five predicted identities. Since some test images may correspond to individuals that do not appear in the training set, the prediction list can also include the special label `new_individual`.

The training set contains 51,033 images and 15,587 individual identities. The test set contains 27,956 images. In addition to individual identity labels, the training data also provides species labels, which are used as auxiliary supervision in our method. Before training, we correct duplicated or misspelled species labels to make the species annotations consistent. For example, `globis` and `pilot_whale` are merged into `short_finned_pilot_whale`, `kiler_whale` is corrected to `killer_whale`, and `bottlenose_dolpin` is corrected to `bottlenose_dolphin`. After this preprocessing step, the dataset contains 26 species classes. Table 1 summarizes the dataset statistics.

Table 1: Summary of the Happywhale dataset used in this project.

Split	# Images	# Identities	# Species
Train	51,033	15,587	26
Test	27,956	-	-

The dataset is highly imbalanced at the identity level. Although there are 15,587 identities in the training set, many identities contain only a small number of images. In particular, 9,258 identities have only one image, and 14,237 identities have no more than five images. The median number of images per identity is only 1, while the most frequent identity contains 400 images. This long-tailed distribution makes the task challenging because the model can easily overfit frequent identities while learning weak representations for rare identities.

The species distribution is also imbalanced after label correction. However, species labels still provide useful coarse-grained information. Therefore, we use species labels in two ways: first, as auxiliary supervision during training, and second, as a species-aware adjustment during retrieval-based prediction.

Figure 2 shows the identity and species distributions. The identity distribution confirms the severe long-tailed property of the dataset, while the species distribution illustrates the imbalance among coarse species categories.

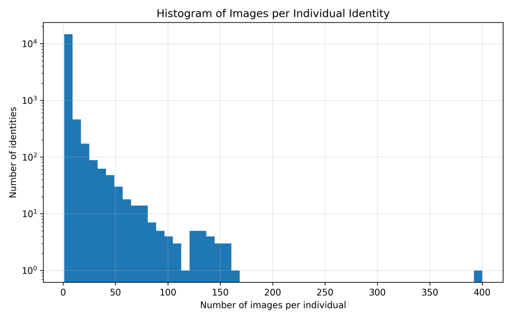
To obtain complementary visual information, we use three types of image views in our pipeline. The raw image preserves the original context and background. The fullbody crop focuses on the visible body region of the animal. The backfin crop emphasizes local fin-related patterns, which are often useful for distinguishing visually similar individuals. Figure 3 shows examples of these three views.

Evaluation Metric. The official evaluation metric is Mean Average Precision at 5 (MAP@5). For each test image, the model predicts a ranked list of five candidate labels. Because each image has only one ground-truth identity, the score of one image depends only on the rank of the correct label. If the correct identity appears at rank 1, the image receives a score of 1.0. If it appears at rank 2, 3, 4, or 5, the score becomes $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, or $\frac{1}{5}$, respectively. If the correct identity does not appear in the top-5 list, the score is 0.

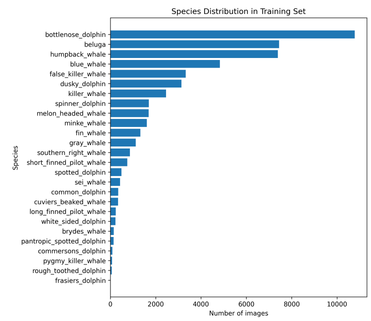
The final MAP@5 is computed by averaging this score over all test images:

$$MAP@5 = \frac{1}{N}\sum_{i=1}^{N} \text{score}_i,$$

where N is the number of test images and score_i is the top-5 ranking score of the i-th image. Repeated predictions of the same label are ignored after the first occurrence.



(a) Identity distribution



(b) Species distribution

Figure 2: Dataset distribution analysis. (a) Long-tailed distribution of individual identities in the training set. Most identities contain only a few images, while a small number of identities contain many samples. In particular, 9,258 identities have only one image, and 14,237 identities have no more than five images. (b) Species distribution after correcting duplicated or misspelled species labels. Species labels provide coarse-grained supervision for the auxiliary species head.

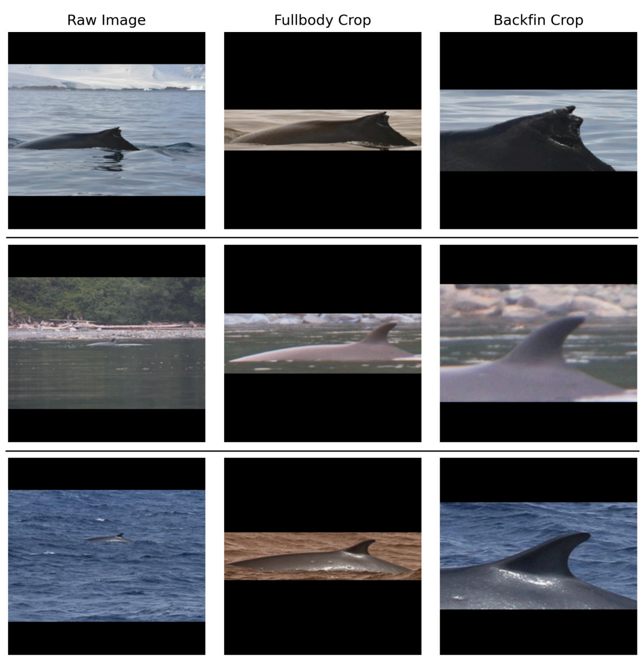


Figure 3: Examples of the three image views used in our pipeline. The raw image preserves the original context, the fullbody crop focuses on the visible animal body, and the backfin crop emphasizes local fin-related cues.

4.2 Implementation Details

We implement four model branches in the final ensemble: EfficientNetV2-XL, EfficientNet-B7, EfficientNet-B6, and ConvNeXt-B. EfficientNet-B7 and ConvNeXt-B follow the original training settings. The EfficientNet-B6 and EfficientNetV2-XL branches are further retrained using pseudo labels generated from our 0.82-score ensemble. In addition, the EfficientNet-B6 branch includes an extra `fullbody_charm` crop view.

Most model branches are trained with multi-view image inputs, including fullbody crops, backfin crops, and raw images. The original training-time crop sampling probabilities are set to 0.70 for fullbody, 0.20 for backfin, and 0.10 for raw images. For the EfficientNet-B6 pseudo-label branch, we additionally use a `fullbody_charm` crop view to provide another body-focused representation.

Each model uses Sub-center ArcFace for identity supervision and an auxiliary species classification head. The identity loss ratio is set to 0.8, and the remaining loss weight is assigned to species supervision. We use separate learning rates for the backbone and classification heads, since the backbone is initialized from pretrained weights while the heads are trained for the Happywhale identity and species labels. Table 2 summarizes the main model branches used in the final ensemble.

Table 2: Training configurations of the model branches used in the final ensemble.

Setting	EfficientNetV2-XL	EfficientNet-B6	EfficientNet-B7	ConvNeXt-B
Backbone	EfficientNetV2-XL	EfficientNet-B6	EfficientNet-B7	ConvNeXt-B
Image size	768×768	1024×1024	1024×1024	1024×1024
Batch size	4	6	6	8
Epochs	40	50	30	30
Backbone LR	3×10^{-5}	1×10^{-5}	2×10^{-5}	1×10^{-5}
Head LR	7×10^{-4}	3×10^{-4}	1×10^{-3}	1×10^{-3}
Weight decay	1×10^{-4}	1×10^{-4}	1×10^{-4}	1×10^{-4}
Identity centers	3	2	3	3
Species centers	3	2	3	3
ArcFace scale	30.0	30.0	30.0	30.0
ArcFace margin	0.30	0.30	0.30	0.30
Identity loss ratio	0.9	0.8	0.8	0.8
Pseudo-label training	two rounds	two rounds	one round	one round

After training, we extract embeddings and logits for both training and test images. During inference, each model uses horizontal flip test-time augmentation. For each model-view pair, we compute three identity score matrices: KNN retrieval score, identity logit score, and single-prototype score. KNN retrieval uses 500 nearest neighbors. The three scores are fused with weights 0.50, 0.25, and 0.25 for KNN, logit, and prototype scores, respectively.

For prototype scoring, one prototype is constructed for each identity by averaging the normalized reference embeddings belonging to that identity, followed by L2 normalization. The query-prototype cosine similarity is then used as the prototype-based identity score. This prototype score provides class-level identity evidence, while KNN provides instance-level nearest-neighbor evidence and the identity logits provide classifier confidence.

We sweep the target `new_individual` ratio over $\{0.165, 0.180, 0.200, 0.215\}$. We also sweep the species bonus over $\{0.00, 0.10\}$ using the top-1 predicted species. The `new_individual` label is treated as an additional candidate with a threshold score, and the threshold is selected by binary search according to the target `new_individual` ratio.

For every single model, we first tune the crop ensemble weights over its available crop views. Then, we combine different model branches using model-level weighted averaging. Table 3 summarizes the final inference settings.

Table 3: Best inference settings for single models and the final ensemble.

Method	Crop weights	Model weight	New ratio	Threshold
EfficientNetV2-XL	0.85 / – / 0.05 / 0.10	–	0.200	0.434760
EfficientNet-B7	0.85 / – / 0.05 / 0.10	–	0.165	0.409352
EfficientNet-B6	0.60 / 0.25 / 0.05 / 0.10	–	0.165	0.455453
ConvNeXt-B	0.85 / – / 0.05 / 0.10	–	0.200	0.325154
Final ensemble	model-specific	0.25 / 0.50 / 0.20 / 0.05	0.180	0.318580

Crop weights are ordered as fullbody / `fullbody_charm` / backfin / raw. The symbol "–" means that the corresponding crop view is not used. Final model weights are ordered as V2-XL / B7 / B6 / ConvNeXt-B.

4.3 Main Results

Table 4 summarizes the final leaderboard results of our main model branches and the final ensemble. Among the single-model branches, EfficientNet-B7 achieves the strongest public and private leaderboard scores. The EfficientNet-B6 and EfficientNetV2-XL branches also provide competitive public scores, and they contribute complementary predictions to the final ensemble.

The final ensemble combines EfficientNetV2-XL, EfficientNet-B7, EfficientNet-B6, and ConvNeXt-B with model weights of 0.25, 0.50, 0.20, and 0.05, respectively. It achieves 0.86095 on the public leaderboard and 0.83044 on the private leaderboard.

Compared with the single-model branches, the ensemble obtains the best public leaderboard score and the best private score among our submitted models. This suggests that different architectures and crop views provide complementary information for whale and dolphin individual identification.

However, the final ensemble still does not surpass the weak baseline on the private leaderboard. This indicates that although model ensemble improves robustness, there remains room for improvement in pseudo-label quality, score calibration, and open-set threshold selection.

Table 4: Main results on the Happywhale leaderboard.

Method	Public	Private
EfficientNet-B7	0.84643	0.81620
EfficientNet-B6	0.83664	0.80298
EfficientNetV2-XL	0.83697	0.79945
ConvNeXt-B	0.80054	0.74866
Final ensemble	0.86095	**0.83044**
Weak baseline	–	0.84909

4.4 Ablation Study

4.4.1 Exploratory Backbone Experiments

Besides the main EfficientNet-B6, EfficientNet-B7, and ConvNeXt-B models, we also explored additional pretrained representation backbones. The goal of these experiments was to examine whether stronger general-purpose or wildlife-pretrained visual representations could improve Happywhale individual identification. Specifically, we tested DINO-based models and a Swin-based wildlife re-identification model.

For the Swin-based experiment, we used MegaDescriptor-L-384, which is a wildlife re-identification pretrained model based on a Swin architecture. This model is related to the universal animal re-identification direction discussed in Section 2.6. Although MegaDescriptor-L-384 is designed for animal re-identification, its input resolution is limited to 384×384. In the Happywhale task, many discriminative cues are very fine-grained, such as small scars, fin edges, body markings, and subtle texture differences. Therefore, the lower input resolution may prevent the model from preserving enough local details. As shown in Table 5, MegaDescriptor-L-384 performs much worse than our EfficientNet-based models.

We also tested DINO-based representations. However, the DINO-related experiments did not provide consistent improvement. Even after tuning the DINO setting with fullbody-only inference, logit-based prediction, a target `new_individual` ratio of 0.165, and 50 training epochs, the result was still limited. Moreover, adding the pretrained DINO branch into the ensemble did not improve the leaderboard score. As shown in Table 5, the DINO-added ensemble achieves 0.80487 public and 0.76859 private scores, which is lower than the EfficientNet-B7 single model.

One possible reason is that DINO learns general-purpose self-supervised representations, while Happywhale requires fine-grained identity-level discrimination. Its feature space may capture species, pose, or global appearance similarity rather than subtle individual cues such as scars, fin edges, and body markings. Moreover, the DINO similarity scores may not be well calibrated with the ArcFace-trained EfficientNet scores.

As a result, directly combining DINO scores with stronger task-specific models can disturb the final ranking instead of improving it.

Overall, these exploratory experiments show that pretrained representation backbones cannot be directly applied to this task without careful adaptation.

For Happywhale individual identification, high-resolution inputs, task-specific metric learning, and calibrated retrieval-based inference are more important than simply using a stronger pretrained representation model.

Table 5: Exploratory backbone and ensemble experiments.

Method	Public	Private
MegaDescriptor-L-384	0.57980	0.64264
DINO tuned setting	0.48660	0.55672
DINO-added ensemble	0.80487	0.76859
EfficientNet-B7	0.84643	0.81612
Final ensemble	0.86095	0.83044

4.4.2 Pseudo-label Training

To analyze the effect of pseudo-label self-training, we compare three EfficientNet-B7 settings. The first setting trains B7 using only the original training data. The second setting resumes training from an epoch-30 checkpoint with pseudo-labeled samples. The third setting trains B7 from scratch using both the original training data and pseudo-labeled samples.

Table 6: Ablation study on pseudo-label training strategies.

Setting	Public LB	Private LB
B7 without pseudo-labels	0.83531	0.80449
B7 pseudo, resume from epoch 30	0.83331	0.80155
B7 pseudo, from scratch	**0.84567**	**0.81535**

The results show that pseudo-label training from scratch improves the B7 model from 0.83531 to 0.84567 on the public leaderboard and from 0.80449 to 0.81535 on the private leaderboard. This indicates that pseudo labels provide useful additional training information.

However, simply resuming training from the epoch-30 checkpoint with pseudo labels does not improve performance. The score decreases to 0.83331 on the public leaderboard and 0.80155 on the private leaderboard. One possible reason is that the embedding space has already become relatively stable after the original training stage. Adding pseudo-labeled samples only during late fine-tuning may disturb the learned embedding structure instead of improving it.

In contrast, training from scratch allows the model to learn original labels and pseudo labels together throughout the full training process. Therefore, the model can integrate pseudo-labeled samples more effectively. This ablation suggests that pseudo-labeling is useful for this task, but it should be introduced from the beginning of training rather than only added during late fine-tuning.

5 Conclusions

In this project, we studied the Happywhale whale and dolphin individual identification task and built an embedding-based open-set retrieval pipeline. Our method combines multi-view image inputs, Sub-center ArcFace identity learning, species auxiliary supervision, KNN-logit-prototype score fusion, threshold-based `new_individual` prediction, and model/crop ensemble.

The final ensemble combines EfficientNetV2-XL, EfficientNet-B7, EfficientNet-B6, and ConvNeXt-B branches. Among our submitted models, the ensemble achieves the best public leaderboard score and the best private score among our own model branches. However, the private score still does not surpass the weak baseline, showing that further improvements are needed in pseudo-label quality, score calibration, and open-set threshold selection.

From this project, we found that data preprocessing is very important for fine-grained animal re-identification. Whether the image preserves useful details, such as scars, fin edges, body markings, and texture differences, strongly affects the final performance. We also found that larger input resolutions generally lead to better results because they preserve more local identity cues. This is especially important for Happywhale, where individual differences may appear only in small regions of the body or fin. Therefore, high-resolution inputs, careful crop selection, metric learning, and calibrated ensemble inference are important for open-set whale and dolphin identification.

Another limitation is the high computational cost of this task. The dataset and intermediate files require more than 80 GB of storage, and high-resolution model training requires substantial GPU memory. In our experiments, at least about 35 GB of GPU memory is needed for efficient training, and one epoch can take around one hour depending on the model and crop setting. These constraints make large-scale hyperparameter search and repeated retraining difficult. Therefore, future work could focus on more efficient training, feature caching, better pseudo-label filtering, and more systematic threshold calibration under limited computational resources.

6 Team Member Contribution

Table 7 summarizes the contribution of each team member to the final project.

Table 7: Team member contribution.

Tasks	**314511037**	**313554063**	**112550028**
Literature survey	34%	33%	33%
Approach design	30%	55%	15%
Approach implementation	40%	40%	20%
Report writing	75%	25%	0%
Slide making and oral presentation	50%	0%	50%

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